STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Thermal equilibrium is transitive.

Q2 - 1 mark

Microscopic energy of molecules and their interactions.

Q3 - 1 mark

Q=DeltaU+W.

Q4 - 1 mark

Q=0.

Q5 - 1 mark

Temperature constant.

Q6 - 1 mark

No.

Q7 - 1 mark

pV=nRT.

Q8 - 1 mark

1-T_c/T_h.

Q9 - 2 marks

Zero boundary work.

Q10 - 2 marks

500 J.

Q11 - 2 marks

50%.

Q12 - 2 marks

Zero.

Q13 - 2 marks

100 J.

Q14 - 2 marks

Isothermal: T constant; adiabatic: Q=0.

Q15 - 3 marks

600 J.

Q16 - 3 marks

40%.

Q17 - 3 marks

W=pDeltaV=2000 J.

Q18 - 3 marks

Second law requires rejected heat or other entropy increase.

Q19 - 4 marks

Reversible: ideal quasi-static frictionless; irreversible: friction or finite-temperature heat flow.

Q20 - 4 marks

Carnot limit 50%; actual efficiency 33.3%; Qout=1000 J.